

Weak Instances of the Two Matrix Code Equivalence Problem

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Abstract. Nowadays, the Matrix Code Equivalence Problem shows potential applicability in constructing efficient and secure advanced digital signatures, focusing on linkable ring signatures, threshold signatures, and blind signatures. Current constructions of these advanced signatures rely on relaxed instantiations of the Matrix Code Equivalence Problem (namely, the 2-MCE problem): given two pairs of equivalent matrix codes, find (if it exists) the secret isometry connecting the pairs. For example, the linkable ring signature construction by Chou et al. (AFRICACRYPT, 2023) builds on top of the Inverse Matrix Code Equivalence Problem: given three equivalent matrix codes, where one pair of the codes is connected by the secret isometry and another by the inverse of that isometry, find the secret isometry. This paper studies the 2-MCE problem, focusing on the family of instances where the secret isometry is (skew) symmetric. Our main contribution corresponds to a polynomial-time algorithm that solves these instances of the 2-MCE problem. Our results have a crucial security impact on the recent blind signature construction proposed by Kuchta, LeGrow, and Persichetti (Cryptography and Communications, 2026), whose security is closely related to the hardness of solving these kinds of instances of the Inverse Matrix Code Equivalent Problem. More precisely, we show that we can break the blind signature construction by Kuchta, LeGrow, and Persichetti (Cryptography and Communications, 2026), with an estimated security of 128 bits, in 33.7 minutes.

Keywords: Cryptanalysis · Kronecker Product · Matrix Code Equivalence · Tensor Product

1 Introduction

In the evolving landscape of post-quantum cryptography, designing efficient and secure cryptographic primitives that resist quantum adversaries remains a central challenge. The cryptographic community is currently increasing its interest in exploring the construction of post-quantum digital signatures that provide advanced functionalities, with a focus on

- Threshold signatures that allow a group of signers to generate a single digital signature collectively;
- Linkable ring signatures that allow a user to sign a message anonymously from within a group (called a “ring”) while also providing a way to publicly identify if two signatures were created by the same individual, without revealing the signer’s identity within the ring.
- Blind signatures that allow a user to obtain a signature on a message without revealing its content to the signer.

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In addition, there is an interesting trend in building these advanced signature schemes on top of equivalence problems (e.g., finding a linear transformation between linear codes, matrix codes, or lattices) — particularly, all relying on relaxed problems where the secret transformation is employed twice [BBN⁺22, CNP⁺23, BBMP24, KTS⁺24, KLP26]. Sadly, current constructions based on the relaxed instantiations of the linear code equivalence (LCE) problem and the lattice isomorphism problem (LIP) raise some security issues when employing the secret transformation twice [BCD⁺24, BCF25]. On the bright side, before this work, the matrix code equivalence looks like a notable candidate for building secure advanced signature schemes — it remains unaffected by the issues spotted in the LCE and LIP scenarios: as long as there is not enough structure on the secrets to be exploited.

The Relaxed Matrix Code Equivalence Problem. An isometry is a homomorphism of matrix codes (i.e., vector subspaces of matrices) that preserves the rank. The Matrix Code Equivalence (MCE) Problem considers finding an isometry between two equivalent matrix codes. The best algorithm to solve the MCE problem corresponds to the recent results of Couvreur and Levrat [CL25], where they present an algorithm to solve the MCE problem with exponential time complexity of $\tilde{O}(q^{k/2})$ field operations in $\mathbb{F}_q$ for a wide family of $(m \times n, k)$ -matrix codes: it restricts to the range of values of k , m , and n such that $m \leq n$, and $k < m^2 - 1$ or $(mn - k) < m^2 - 1$. The Two Matrix Code Equivalence Problem (2-MCE) is a relaxed variant of the MCE problem and asks to determine the secret isometry connecting two given pairs of equivalent matrix codes. To our knowledge, the best approach to solving the 2-MCE problem is to solve the original MCE problem without taking advantage of knowledge of two applications of the same secret isometry.

1.1 Motivation

Chou et al. describe in [CNP⁺23] a linkable ring signature scheme based on the hardness of the so-called Inverse Matrix Code Equivalence (IMCE) Problem, which asks to retrieve the secret isometry connecting three equivalent matrix codes (one pair of codes is connected by the secret isometry, while another by the inverse of that isometry). Subsequently, Battagliola et al. present in [BBMP24] a threshold signature based on the Two Matrix Code Equivalence Problem. Recently, Kuchta, LeGrow, and Persichetti introduce in [KLP26] a novel blind signature scheme based on a new relaxed computational assumption: the Modified Inverse Matrix Code Equivalence (MIMCE) Problem, where the secret isometry is (skew) symmetric, and the task corresponds to finding the square of such an isometry. The authors argue that using the matrix code equivalent setting, where the attack from [BCD⁺24] does not apply, avoids the pitfalls associated with linear codes when using twice the secret. However, despite the recent cryptanalysis from [CL25, GMPT26], the authors of [KLP26] make a conservative choice and propose sets of parameters with an estimated complexity of 128-bit security.

As an interesting observation, depending on the required advanced functionality of the signature, the current state-of-the-art constructions impose some additional properties on the secret isometry. For example, the Relaxed Matrix Code Equivalence Problem from [KLP26] is a special instantiation of IMCE, and the IMCE problem determines a particular instantiation of the 2-MCE problem. However, restricting the secret isometry in this way may inadvertently reduce security. This motivates our central question:

Given an instance of 2-MCE where the secret isometry is constrained to a special family, does this restriction make the problem easier to solve?

We answer this question in the affirmative for a large class of cases, with particular focus on the scenario where the isometries are (skew) symmetric as instantiated in [KLP26]. Specifically, we show that these structured versions are polynomial-time solvable for any possible parameter triples (k, m, n) .

1.2 Technical Overview

Let k , m , and n be three positive integers, and q a power of a prime number. Informally, given three k -dimensional vector subspaces of $\mathbb{F}_q^{mn}$ generated by $\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3 \in \mathbb{F}_q^{k \times mn}$, the Two Matrix Code Equivalence Problem asks to find (if they exist) three invertible matrices $\mathbf{S}$, $\mathbf{A}$, and $\mathbf{B}$ defined over the finite field $\mathbb{F}_q$ such that $\mathbf{G}_1 = \mathbf{S}\mathbf{G}_0\mathbf{Q}$ and $\mathbf{G}_3 = \mathbf{S}\mathbf{G}_2\mathbf{Q}$ where $\mathbf{Q} = \mathbf{A}^\top \otimes \mathbf{B}$. This work focuses on the scenario where the matrices $\mathbf{A}$ and $\mathbf{B}$ are (skew) symmetric, and presents a new approach to recovering such matrices $\mathbf{S}$, $\mathbf{A}$, and $\mathbf{B}$. We describe the main steps of our algorithm in Figure 1.

Algorithm: Solver for the 2-sMCE problem

Input: An instance $\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3 \in \mathbb{F}_q^{k \times mn}$ of the 2-sMCE problem

Output: Invertible matrices $\mathbf{S}, \mathbf{A}$ and $\mathbf{B}$ such that $\mathbf{G}_1 = \mathbf{S}\mathbf{G}_0\mathbf{Q}$ and $\mathbf{G}_3 = \mathbf{S}\mathbf{G}_2\mathbf{Q}$ where $\mathbf{Q} = \mathbf{A}^\top \otimes \mathbf{B}$

1. Compute

$$\begin{aligned} \mathbf{X}_1 &= (\mathbf{G}_0\mathbf{G}_1^\top)(\mathbf{G}_2\mathbf{G}_1^\top)^{-1}, & \mathbf{Y}_1 &= (\mathbf{G}_1\mathbf{G}_0^\top)(\mathbf{G}_3\mathbf{G}_0^\top)^{-1}, \\ \mathbf{X}_2 &= (\mathbf{G}_2\mathbf{G}_3^\top)(\mathbf{G}_2\mathbf{G}_1^\top)^{-1}, & \mathbf{Y}_2 &= (\mathbf{G}_3\mathbf{G}_2^\top)(\mathbf{G}_3\mathbf{G}_0^\top)^{-1}, \end{aligned}$$

and find a nonzero solution $\mathbf{x} = [\mathbf{S}_{1,1} \ \mathbf{S}_{1,2} \ \dots \ \mathbf{S}_{k,k-1} \ \mathbf{S}_{k,k}]$ of the linear system

$$\begin{bmatrix} \mathbf{I}_k \otimes \mathbf{X}_1^\top - \mathbf{Y}_1 \otimes \mathbf{I}_k \\ \mathbf{I}_k \otimes \mathbf{X}_2^\top - \mathbf{Y}_2 \otimes \mathbf{I}_k \end{bmatrix} \mathbf{x} = \mathbf{0}.$$

2. Solve the linear system $(\mathbf{S}^{-1}\mathbf{G}_1)(\mathbf{X} \otimes \mathbf{I}_k) = \mathbf{G}_0(\mathbf{I}_k \otimes \mathbf{Y})$ for the variables $\mathbf{X} = \mathbf{A}^{-\top}$ and $\mathbf{Y} = \mathbf{B}$.

3. **Return** $(\mathbf{S}, \mathbf{A}, \mathbf{B})$

Figure 1: We consider integers $nm > k$, and prime power numbers q .

Since each calculation in the algorithm presented in Figure 1 includes matrix multiplications and linear system resolutions, we find that our proposed algorithm runs in polynomial time on the variables k, m , and n . We discuss in section Section 3 the correctness of the algorithm, and for which parameter tuples (k, m, n) it correctly recovers the secret matrices $\mathbf{S}, \mathbf{A}$ and $\mathbf{B}$ of random 2-sMCE instances.

Remark 1. We note that the algorithm in Figure 1 requires a rank-maximality condition on the matrices $\mathbf{G}_0\mathbf{G}_1^\top$, $\mathbf{G}_2\mathbf{G}_3^\top$, $\mathbf{G}_0\mathbf{G}_3^\top$ and $\mathbf{G}_1\mathbf{G}_2^\top$. We emphasize that this condition is expected to be satisfied for random instances of the 2-sMCE problem of cryptographic interest. However, if this condition is not satisfied for a given instance, we suggest translating the input 2-sMCE instance into another instance $\mathbf{G}'_0, \mathbf{G}'_1, \mathbf{G}'_2, \mathbf{G}'_3 \in \mathbb{F}_q^{k \times mn}$ that satisfies the required rank-maximality condition and keep the same $\mathbf{S}$, and replacing $\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3$ with $\mathbf{G}'_0, \mathbf{G}'_1, \mathbf{G}'_2, \mathbf{G}'_3$ on the Line 1 of Figure 1. For example, one approach would be as follows.

1. Sample random invertible symmetric matrices $\mathbf{L}_A, \mathbf{R}_A, \mathbf{L}_B$ and $\mathbf{R}_B$
2. Compute $\mathbf{L}_Q = \mathbf{L}_A^\top \otimes \mathbf{L}_B$ and $\mathbf{R}_Q = \mathbf{R}_A^\top \otimes \mathbf{R}_B$.
3. Calculate $\mathbf{G}'_0 = \mathbf{G}_0(\mathbf{L}_Q^{-1})$, $\mathbf{G}'_1 = \mathbf{G}_1\mathbf{R}_Q$, $\mathbf{G}'_2 = \mathbf{G}_2(\mathbf{L}_Q^{-1})$ and $\mathbf{G}'_3 = \mathbf{G}_3\mathbf{R}_Q$.

Outline. Section 2 introduces the required mathematical definitions and theory concerning matrices, tensors and matrix codes. In particular, Section 2.1 focuses on the properties of tensor products and Section 2.2 describes the Relaxed Matrix Code Equivalence Problems analyzed in this work. In addition, Section 2.3 presents new definitions and results that link Matrix Code Equivalence instances with the dual of the matrix codes, which could be of independent interest. Section 3 details the main results on the Two Matrix Code Equivalence Problem when the secret isometry is (skew) symmetric, and Section 4 discuss the cryptographic implications on [KLP26].

2 Preliminaries

Let q be a small odd prime number and let $\mathbb{F}_q$ be a finite field that has q elements. We denote the group of invertible $n \times n$ matrices with entries in $\mathbb{F}_q$ by $\text{GL}_n(\mathbb{F}_q)$, and the set of invertible symmetric and skew-symmetric matrices by $\text{S}_k(\mathbb{F}_q)$. We define $\mathbb{M}_r^{m \times n}(\mathbb{F}_q)$ as the set of m -by- n matrices of rank r , that is, $\mathbb{M}_r^{m \times n}(\mathbb{F}_q) = \{\mathbf{M} \in \mathbb{F}_q^{m \times n} : \text{rank}(\mathbf{M}) = r\}$. If $r = m$, we write $\mathbb{M}^{m \times n}(\mathbb{F}_q)$ to mean $\mathbb{M}_m^{m \times n}(\mathbb{F}_q)$. For any matrix $\mathbf{M} \in \text{GL}_n(\mathbb{F}_q)$, we write $\mathbf{M}^{-\top}$ to mean the inverse of the matrix $\mathbf{M}^\top$. We denote the Kronecker product of two matrices $\mathbf{A} \in \mathbb{F}_q^{n \times m}$ and $\mathbf{B} \in \mathbb{F}_q^{r \times s}$ by $\mathbf{A} \otimes \mathbf{B} \in \mathbb{F}_q^{nr \times ms}$. We denote the n -by- n identity matrix by $\mathbf{I}_n$.

2.1 The Kronecker Product and Some Useful Properties

For any matrix $\mathbf{M} \in \mathbb{F}_q^{m \times n}$, $\text{vec}(\mathbf{M})$ denotes the column vector of mn coefficients consisting of the concatenation of the rows of the matrix $\mathbf{M}$. We emphasize that the Kronecker product is a tensor product between matrices viewed as vector spaces. We highlight some useful properties below.

- For any two m -by- n matrices $\mathbf{M}$ and $\mathbf{N}$, and two n -by- r matrices $\mathbf{S}$ and $\mathbf{R}$, it holds that $(\mathbf{M} \otimes \mathbf{N})(\mathbf{R} \otimes \mathbf{S}) = (\mathbf{M}\mathbf{S} \otimes \mathbf{N}\mathbf{R})$.
- For any two matrices $\mathbf{M}$ and $\mathbf{N}$, it holds that $(\mathbf{M} \otimes \mathbf{N})^\top = \mathbf{M}^\top \otimes \mathbf{N}^\top$.
- For any two matrices $\mathbf{M}$ and $\mathbf{N}$ and constant a , $a(\mathbf{M} \otimes \mathbf{N}) = (a\mathbf{M}) \otimes \mathbf{N} = \mathbf{M} \otimes (a\mathbf{N})$ holds.
- For any two invertible matrices $\mathbf{M}$ and $\mathbf{N}$, $(\mathbf{M} \otimes \mathbf{N})^{-1} = \mathbf{M}^{-1} \otimes \mathbf{N}^{-1}$ holds.
- For any two matrices $\mathbf{M}$ and $\mathbf{N}$, we have $\text{rank}(\mathbf{M} \otimes \mathbf{N}) = \text{rank}(\mathbf{M})\text{rank}(\mathbf{N})$.
- For any three matrices $\mathbf{M} \in \mathbb{F}_q^{m \times n}$, $\mathbf{N} \in \mathbb{F}_q^{s \times t}$ and $\mathbf{X} \in \mathbb{F}_q^{n \times t}$, we have $[\mathbf{M} \otimes \mathbf{N}] \text{vec}(\mathbf{X}) = \text{vec}(\mathbf{M}\mathbf{X}\mathbf{N}^\top)$.

Given a matrix $\mathbf{M} \in \mathbb{F}_q^{m \times n}$, the commutation matrix $\mathbf{K}_{m,n}$ is the mn -by- mn permutation matrix that satisfies $\mathbf{K}_{m,n} \text{vec}(\mathbf{M}) = \text{vec}(\mathbf{M}^\top)$. In particular, for any pairs of matrices $\mathbf{A} \in \mathbb{F}_q^{m \times n}$ and $\mathbf{B} \in \mathbb{F}_q^{r \times s}$, the commutation matrix relates $\text{vec}(\mathbf{A} \otimes \mathbf{B})$ with $\text{vec}(\mathbf{A}) \otimes \text{vec}(\mathbf{B})$ as follows:

$$\text{vec}(\mathbf{A} \otimes \mathbf{B}) = (\mathbf{I}_n \otimes \mathbf{K}_{s,m} \otimes \mathbf{I}_r)(\text{vec}(\mathbf{A}) \otimes \text{vec}(\mathbf{B})).$$

Moreover, the commutation matrices $\mathbf{K}_{m,r}$ and $\mathbf{K}_{n,s}$ link the matrices $\mathbf{A} \otimes \mathbf{B}$ and $\mathbf{B} \otimes \mathbf{A}$ by equality $\mathbf{A} \otimes \mathbf{B} = \mathbf{K}_{m,r} \mathbf{B} \otimes \mathbf{A} (\mathbf{K}_{n,s})^\top$.

2.2 Matrix Code Equivalence Problems

A $(m \times n, k)$ *matrix code* is a subspace $\mathcal{D}$ of dimension k of the space of $m \times n$ matrices. We say that two matrix codes $\mathcal{D}, \mathcal{D}'$ are *equivalent* if there exist two matrices $\mathbf{A} \in \text{GL}_m(\mathbb{F}_q)$ and $\mathbf{B} \in \text{GL}_n(\mathbb{F}_q)$ such that $\mathcal{D}' = \mathbf{A}\mathcal{D}\mathbf{B}$. Each $(m \times n, k)$ *matrix code* $\mathcal{D}$ is isomorphic to the k dimensional $\mathbb{F}_q$ subspace $\text{vec}(\mathcal{D}) = \{\text{vec}(\mathbf{D}) \mid \mathbf{D} \in \mathcal{D}\}$ of $\mathbb{F}_q^{mn}$ [CNP⁺23]. We call $\mathbf{G} \in \mathbb{F}_q^{k \times mn}$ a *generator matrix* of $\mathcal{D}$ if its rows form a basis of $\text{vec}(\mathcal{D})$. In particular, we focus on the definition of the MCE and IMCE problems from [CNP⁺23] given in terms of the Kronecker product.

Definition 1 (Matrix Code Equivalence (MCE) Problem). Let $\mathbf{G}_0, \mathbf{G}_1 \in \mathbb{F}_q^{k \times mn}$ be generator matrices of two equivalent $(m \times n, k)$ -matrix codes $\mathcal{D}_0, \mathcal{D}_1$ respectively. The Matrix Code Equivalence problem asks to find $\mathbf{S} \in \text{GL}_k(\mathbb{F}_q)$, $\mathbf{A} \in \text{GL}_m(\mathbb{F}_q)$ and $\mathbf{B} \in \text{GL}_n(\mathbb{F}_q)$ such that $\mathbf{G}_1 = \mathbf{S}\mathbf{G}_0(\mathbf{A}^\top \otimes \mathbf{B})$.

Definition 2 (Inverse Matrix Code Equivalence (IMCE) Problem). Let $\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2 \in \mathbb{F}_q^{k \times mn}$ be generators of three $(m \times n, k)$ -matrix codes $\mathcal{D}_0, \mathcal{D}_1$ and $\mathcal{D}_2$ respectively. The Inverse Matrix Code Equivalence problem is to find (if they exist) $\mathbf{S} \in \text{GL}_k(\mathbb{F}_q)$, $\mathbf{A} \in \text{GL}_m(\mathbb{F}_q)$ and $\mathbf{B} \in \text{GL}_n(\mathbb{F}_q)$ such that $\mathbf{G}_1 = \mathbf{S}\mathbf{G}_0\mathbf{Q}$ and $\mathbf{G}_2 = \mathbf{S}^{-1}\mathbf{G}_0\mathbf{Q}^{-1}$ with $\mathbf{Q} = (\mathbf{A}^\top \otimes \mathbf{B}) \in \text{GL}_{mn}(\mathbb{F}_q)$.

In the context of blind signatures, the recent construction from [KLP26] introduces a modified variant of the IMCE problem, which we describe in terms of the tensor product below.

Definition 3 (A Modified Inverse Matrix Code Equivalence (MIMCE) Problem). Let $\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2 \in \mathbb{F}_q^{k \times mn}$ be generators of three $(m \times n, k)$ -matrix codes $\mathcal{D}_0, \mathcal{D}_1$ and $\mathcal{D}_2$, respectively, that satisfy $\mathbf{G}_1 = \mathbf{S}\mathbf{G}_0\mathbf{Q}$ and $\mathbf{G}_2 = \mathbf{S}^{-1}\mathbf{G}_0\mathbf{Q}^{-1}$ with $\mathbf{Q} = (\mathbf{A}^\top \otimes \mathbf{B}) \in \text{GL}_{mn}(\mathbb{F}_q)$ for some secret matrices $\mathbf{S} \in \text{GL}_k(\mathbb{F}_q)$, $\mathbf{A} \in \mathbb{S}_m(\mathbb{F}_q)$ and $\mathbf{B} \in \mathbb{S}_n(\mathbb{F}_q)$. The Modified Inverse Matrix Code Equivalence problem asks to find $\mathbf{S}' \in \text{GL}_k(\mathbb{F}_q)$, $\mathbf{C} \in \mathbb{S}_m(\mathbb{F}_q)$ and $\mathbf{D} \in \mathbb{S}_n(\mathbb{F}_q)$ such that $\mathbf{G}_1 = \mathbf{S}'\mathbf{G}_2\mathbf{Q}'$ with $\mathbf{Q}' = (\mathbf{C}^\top \otimes \mathbf{D}) \in \text{GL}_{mn}(\mathbb{F}_q)$.

Remark 2. Notice that MIMCE asks to find the “square” of the secret isometry, and defines an IMCE instance; then it becomes natural to focus on the 2-MCE instances with the secrets being (skew) symmetric since IMCE is a special case of 2-MCE. In other words, we try to be as general as possible and center our analysis on the 2-MCE problem, which could be employed to build (and improve) threshold and linakble ring signatures.

2.3 Additional Definitions

Let $\mathcal{D}$ be a $(m \times n, k)$ matrix code and $\mathbf{G} \in \mathbb{F}_q^{k \times mn}$ be the generator matrix of $\mathcal{D}$. We define the *rate* of the code $\mathcal{D}$ as the quantity $\delta = \frac{k}{mn}$. Furthermore, we define the parity-check matrix of $\mathcal{D}$ as the matrix $\mathbf{H} \in \mathbb{F}_q^{(mn-k) \times mn}$ such that $\mathbf{H}\mathbf{d} = \mathbf{0}$ for each $\mathbf{d} \in \text{vec}(\mathcal{D})$. The parity-check matrix generates the dual code of $\text{vec}(\mathcal{D})$, and thus we define the dual code of $\mathcal{D}$ as the $(m \times n, mn - k)$ matrix code $\mathcal{D}^\perp$ isomorphic to the code generated by the parity check matrix $\mathbf{H}$. We present in Lemma 1 and Corollary 1 the relation between the parity-check matrices of two equivalent matrix codes.

Lemma 1. Let $\mathbf{G}_0, \mathbf{G}_1 \in \mathbb{F}_q^{k \times mr}$ be generator matrices of two equivalent $(m \times r, k)$ -matrix codes $\mathcal{D}_0, \mathcal{D}_1$ respectively; that is, there exist matrices $\mathbf{S} \in \text{GL}_k(\mathbb{F}_q)$, $\mathbf{A} \in \text{GL}_m(\mathbb{F}_q)$ and $\mathbf{B} \in \text{GL}_r(\mathbb{F}_q)$ such that $\mathbf{G}_1 = \mathbf{S}\mathbf{G}_0(\mathbf{A}^\top \otimes \mathbf{B})$. Let $\mathbf{H}_0, \mathbf{H}_1$ be the parity-check matrices of $\mathcal{D}_0, \mathcal{D}_1$ respectively. Then, $\mathbf{H}_1 = \mathbf{R}\mathbf{H}_0(\mathbf{A}^{-1} \otimes \mathbf{B}^{-\top})$ holds for some $\mathbf{R} \in \text{GL}_{mn-k}(\mathbb{F}_q)$.

Proof. By definition, the parity-check matrices $\mathbf{H}_0$ and $\mathbf{H}_1$ satisfy $\mathbf{H}_0\mathbf{G}_0^\top = \mathbf{0}$ and $\mathbf{H}_1\mathbf{G}_1^\top = \mathbf{0}$. Notice that the matrix $\mathbf{H} = \mathbf{H}_0(\mathbf{A}^{-\top} \otimes \mathbf{B}^{-1}) \in \mathbb{F}_q^{(mn-k) \times mn}$ has rank

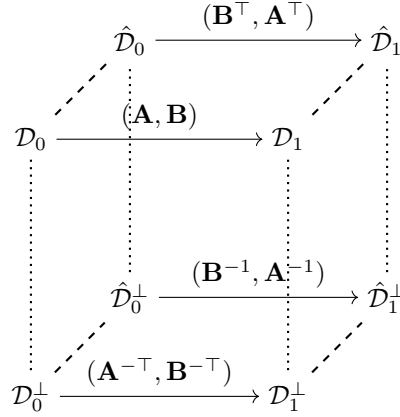


Figure 2: Code Equivalence Diagram determined by Corollaries 1 and 2: $\mathcal{D}_0, \mathcal{D}_1$ are two equivalent $(m \times n, k)$ -matrix codes; that is, there exist two matrices $\mathbf{A} \in \text{GL}_m(\mathbb{F}_q)$ and $\mathbf{B} \in \text{GL}_n(\mathbb{F}_q)$ such that $\mathcal{D}_1 = \mathbf{A}\mathcal{D}_0\mathbf{B}$. Here, $\hat{\mathcal{D}}_0$ and $\hat{\mathcal{D}}_1$ denote the $(n \times m, k)$ -matrix codes from Corollary 2. In addition, $\mathcal{D}_0^\perp$ and $\mathcal{D}_1^\perp$ determine the $(m \times n, k)$ -matrix dual codes of $\mathcal{D}_0$ and $\mathcal{D}_1$, while $\hat{\mathcal{D}}_0^\perp$ and $\hat{\mathcal{D}}_1^\perp$ the $(n \times m, k)$ -matrix dual codes of $\hat{\mathcal{D}}_0$ and $\hat{\mathcal{D}}_1$. The horizontal edges describe the code equivalence determined by specific pairs of matrices.

$(mn - k)$, and

$$\begin{aligned}
 \mathbf{H}\mathbf{G}_1^\top &= (\mathbf{H}_0(\mathbf{A}^{-1} \otimes \mathbf{B}^{-\top}))(\mathbf{S}\mathbf{G}_0(\mathbf{A}^\top \otimes \mathbf{B}))^\top \\
 &= (\mathbf{H}_0(\mathbf{A}^{-1} \otimes \mathbf{B}^{-\top}))((\mathbf{A} \otimes \mathbf{B}^\top)\mathbf{G}_0^\top\mathbf{S}^\top) \\
 &= \mathbf{H}_0(\mathbf{I}_m \otimes \mathbf{I}_n)\mathbf{G}_0^\top\mathbf{S}^\top = \mathbf{H}_0\mathbf{I}_{mn}\mathbf{G}_0^\top\mathbf{S}^\top \\
 &= \mathbf{H}_0\mathbf{G}_0^\top\mathbf{S}^\top = \mathbf{0}.
 \end{aligned}$$

In other words, $\mathbf{H}$ is a generator matrix of dual code $\mathcal{D}_1^\perp$. Hence, there exists $\mathbf{R} \in \text{GL}_{mn-k}(\mathbb{F}_q)$ such that $\mathbf{H}_1 = \mathbf{R}\mathbf{H}_0(\mathbf{A}^{-1} \otimes \mathbf{B}^{-\top})$. $\square$

Corollary 1. *Let $\mathcal{D}_0, \mathcal{D}_1$ be two equivalent $(m \times n, k)$ -matrix codes; that is, there exist two matrices $\mathbf{A} \in \text{GL}_m(\mathbb{F}_q)$ and $\mathbf{B} \in \text{GL}_n(\mathbb{F}_q)$ such that $\mathcal{D}_1 = \mathbf{A}\mathcal{D}_0\mathbf{B}$. Then, $\mathcal{D}_1^\perp = \mathbf{A}^{-\top}\mathcal{D}_0^\perp\mathbf{B}^{-\top}$.*

Proof. The proof follows from the equivalence given by Lemma 1, which is shown in terms of Kronecker product. $\square$

We highlight that Corollary 1 illustrates a different result from [CL25, Lemma 2] since our definition of the dual code focuses on the generator matrix, which differs from the definition in [CL25]. For example, [CL25, Lemma 2] allows us to swap the usage of the matrices $\mathbf{A}$ and $\mathbf{B}$ in the code equivalence relation between two $(n \times m, k)$ -matrix codes $\tilde{\mathcal{D}}_0$ and $\tilde{\mathcal{D}}_1$, more precisely, $\tilde{\mathcal{D}}_1 = \mathbf{B}^{-\top}\tilde{\mathcal{D}}_0\mathbf{A}^{-\top}$. In the tensor product notation, this means an equivalence determined by $\mathbf{B}^{-1} \otimes \mathbf{A}^{-\top}$ instead of $\mathbf{A}^\top \otimes \mathbf{B}$. We show in Corollary 2 which $(n \times m, k)$ -matrix codes allow a result similar to [CL25, Lemma 2], that is, that swaps the usage of the matrices $\mathbf{A}$ and $\mathbf{B}$ in the code equivalence. Figure 2 summarizes the code equivalences that Corollaries 1 and 2 imply.

Corollary 2. *Let $\mathcal{D}_0, \mathcal{D}_1$ be two equivalent $(m \times n, k)$ -matrix codes; that is, there exist two matrices $\mathbf{A} \in \text{GL}_m(\mathbb{F}_q)$ and $\mathbf{B} \in \text{GL}_n(\mathbb{F}_q)$ such that $\mathcal{D}_1 = \mathbf{A}\mathcal{D}_0\mathbf{B}$. Let $\mathbf{G}_0$ and $\mathbf{G}_1 \in \mathbb{F}_q^{k \times mr}$ be the generator matrices of $\mathcal{D}_0$ and $\mathcal{D}_1$, respectively. Let $\hat{\mathbf{G}}_i = \mathbf{G}_i\mathbf{K}_{m,n}$ where $\mathbf{K}_{m,n}$*

denotes the commutation matrix and $i \in \{0, 1\}$. Let $\hat{\mathcal{D}}_0$ and $\hat{\mathcal{D}}_1$ be the $(n \times m, k)$ -matrix codes with generator matrices $\hat{\mathbf{G}}_0$ and $\hat{\mathbf{G}}_1$, respectively. Then, $\hat{\mathcal{D}}_1 = \mathbf{B}^\top \hat{\mathcal{D}}_0 \mathbf{A}^\top$.

Proof. Recall that $\mathbf{A}^\top \otimes \mathbf{B} = \mathbf{K}_{m,n} \mathbf{B} \otimes \mathbf{A}^\top (\mathbf{K}_{m,n})^\top$ holds, where $\mathbf{K}_{m,n}$ denotes the commutation matrix. Since $\mathbf{G}_1 = \mathbf{S} \mathbf{G}_0 (\mathbf{A}^\top \otimes \mathbf{B})$ for some matrix $\mathbf{S} \in \text{GL}_k(\mathbb{F}_q)$, we find that $\mathbf{G}_1 \mathbf{K}_{m,n} = \mathbf{S} (\mathbf{G}_0 \mathbf{K}_{m,n}) (\mathbf{B} \otimes \mathbf{A}^\top)$ holds. Hence, $\hat{\mathbf{G}}_1 = \mathbf{S} \hat{\mathbf{G}}_0 (\mathbf{B} \otimes \mathbf{A}^\top)$, which is equivalent to $\hat{\mathcal{D}}_1 = \mathbf{B}^\top \hat{\mathcal{D}}_0 \mathbf{A}^\top$. $\square$

3 On Weak Instances of the 2-MCE problem

This section focuses on a specific family of instances of the 2-MCE problem where the secret matrices A and B are (skew) symmetric. We label these instances as 2-sMCE to highlight this (skew) symmetric scenario. Thus, let us assume that we have an instance $(\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2)$ of the 2-sMCE problem as formalized in Definition 4.

Definition 4 (Symmetric Two Matrix Code Equivalence (2-sMCE) Problem). Let $\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3 \in \mathbb{F}_q^{k \times mn}$ be generators of three $(m \times n, k)$ -matrix codes $\mathcal{D}_0, \mathcal{D}_1, \mathcal{D}_2$ and $\mathcal{D}_3$ respectively. The Two Inverse Matrix Code Equivalence problem is to find (if they exist) $\mathbf{S} \in \text{GL}_k(\mathbb{F}_q)$, $\mathbf{A} \in \mathbb{S}_m(\mathbb{F}_q)$ and $\mathbf{B} \in \mathbb{S}_n(\mathbb{F}_q)$ such that $\mathbf{G}_1 = \mathbf{S} \mathbf{G}_0 \mathbf{Q}$ and $\mathbf{G}_3 = \mathbf{S} \mathbf{G}_2 \mathbf{Q}$ with $\mathbf{Q} = (\mathbf{A}^\top \otimes \mathbf{B}) \in \text{GL}_{mn}(\mathbb{F}_q)$.

We discuss the correctness of our algorithm in Figure 1 below.

3.1 On the Details of Step 1 in Figure 1

Lemma 2. Consider an instance $\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3 \in \mathbb{F}_q^{k \times mn}$ of the 2-sMCE problem. Then,

$$\mathbf{S}(\mathbf{G}_0 \mathbf{G}_1^\top) = \epsilon(\mathbf{G}_1 \mathbf{G}_0^\top) \mathbf{S}^\top \quad (1)$$

$$\mathbf{S}(\mathbf{G}_2 \mathbf{G}_3^\top) = \epsilon(\mathbf{G}_3 \mathbf{G}_2^\top) \mathbf{S}^\top \quad (2)$$

$$\mathbf{S}(\mathbf{G}_0 \mathbf{G}_3^\top) = \epsilon(\mathbf{G}_1 \mathbf{G}_2^\top) \mathbf{S}^\top \quad (3)$$

where

$$\epsilon = \begin{cases} 1 & \text{if } \mathbf{Q} = \mathbf{Q}^\top \\ -1 & \text{otherwise.} \end{cases}$$

Proof. We know that $\mathbf{Q}^\top = \epsilon \mathbf{Q}$ for some $\epsilon \in \{1, -1\}$. For each pair $(i, j) \in \{(0, 1), (2, 3)\}$, we find that

$$\mathbf{G}_i \mathbf{G}_j^\top = (\mathbf{S}^{-1} \mathbf{G}_i \mathbf{Q}^{-1})(\mathbf{S} \mathbf{G}_j \mathbf{Q})^\top = \epsilon \mathbf{S}^{-1} (\mathbf{G}_j \mathbf{G}_i^\top) \mathbf{S}^\top.$$

Similarly, we obtain

$$\mathbf{G}_0 \mathbf{G}_3^\top = (\mathbf{S}^{-1} \mathbf{G}_1 \mathbf{Q}^{-1})(\mathbf{S} \mathbf{G}_2 \mathbf{Q})^\top = \epsilon \mathbf{S}^{-1} (\mathbf{G}_1 \mathbf{G}_2^\top) \mathbf{S}^\top$$

Therefore, it follows that Equations (1) to (3) hold. $\square$

Notice that Equations (1) and (2) have an inherent (skew) symmetric structure that limits the rank of the systems to $\frac{1}{2}k(k-1) - \epsilon$, while Equation (3) does not. As a way to “break” the (skew) symmetric structure in the systems, we propose to analyze the linear systems determined by Equation (1) (resp. Equation (2)) and Equation (3). We provide in Propositions 1 and 2 a detailed analysis of the rank in both scenarios.

Proposition 1. Consider an instance $\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3 \in \mathbb{F}_q^{k \times mn}$ of the 2-sMCE problem such that $\mathbf{G}_0 \mathbf{G}_1^\top, \mathbf{G}_0 \mathbf{G}_3^\top, \mathbf{G}_1 \mathbf{G}_2^\top \in \mathbb{F}_q^{k \times k}$ are full rank matrices. Write $\mathbf{X} = (\mathbf{G}_0 \mathbf{G}_1^\top)(\mathbf{G}_2 \mathbf{G}_1^\top)^{-1}$ and $\mathbf{Y} = (\mathbf{G}_1 \mathbf{G}_0^\top)(\mathbf{G}_3 \mathbf{G}_0^\top)^{-1}$. Then

$$\mathbf{S}\mathbf{X} = \mathbf{Y}\mathbf{S} \quad (4)$$

holds. Moreover, $\mathbf{x} = \text{vec}(\mathbf{S})$ is a nonzero solution of the linear system $\mathbf{M}\mathbf{x} = \mathbf{0}$ where

$$\mathbf{M} = [\mathbf{I}_k \otimes \mathbf{X}^\top - \mathbf{Y} \otimes \mathbf{I}_k]$$

has a rank at most equal to $k^2 - k$.

Proof. From Equation (1), we find that $\mathbf{S}^\top = \epsilon(\mathbf{G}_3 \mathbf{G}_0^\top)^{-1} \mathbf{S}(\mathbf{G}_2 \mathbf{G}_1^\top)$ holds. Observe that plugging this last equality into Equation (2) gives

$$\begin{aligned} \mathbf{S}(\mathbf{G}_0 \mathbf{G}_1^\top) &= \epsilon(\mathbf{G}_1 \mathbf{G}_0^\top) \mathbf{S}^\top \\ &= \epsilon(\mathbf{G}_1 \mathbf{G}_0^\top)(\epsilon(\mathbf{G}_3 \mathbf{G}_0^\top)^{-1} \mathbf{S}(\mathbf{G}_2 \mathbf{G}_1^\top)) \\ &= \epsilon^2(\mathbf{G}_1 \mathbf{G}_0^\top)(\mathbf{G}_3 \mathbf{G}_0^\top)^{-1} \mathbf{S}(\mathbf{G}_2 \mathbf{G}_1^\top) \end{aligned}$$

Since $\epsilon \in \{1, -1\}$, we see that $\epsilon^2 = 1$. Notice that if we multiply the above equation by $(\mathbf{G}_2 \mathbf{G}_1^\top)^{-1}$ on the right, we obtain Equation (4). From the tensor properties in Section 2.1, we verify that $[\mathbf{I}_k \otimes \mathbf{X}^\top - \mathbf{Y} \otimes \mathbf{I}_k] \text{vec}(\mathbf{S}) = \mathbf{0}$. Now, for any element $\mathbf{C}$ in the centralizer $\mathcal{C}(\mathbf{Y}) := \{\mathbf{M} \in \mathbb{F}_q^{k \times k} : \mathbf{M}\mathbf{Y} = \mathbf{Y}\mathbf{M}\}$ of $\mathbf{Y}$, the matrix $\mathbf{S}' = \mathbf{C}\mathbf{S}$ satisfies

$$\mathbf{Y}\mathbf{S}' = \mathbf{X}\mathbf{C}\mathbf{S} = \mathbf{C}\mathbf{Y}\mathbf{S} = \mathbf{C}\mathbf{S}\mathbf{X} = \mathbf{S}'\mathbf{X}.$$

That is, each element of the lateral class $\mathcal{C}(\mathbf{Y})\mathbf{S} = \{\mathbf{C}\mathbf{S} : \mathbf{C} \in \mathcal{C}(\mathbf{Y})\}$ is a solution of Equation (4). Since $\mathbf{Y}$ has rank k and $\mathbf{S}$ is invertible, $\mathcal{C}(\mathbf{Y})$ is a vector space of dimension at least k [HJ94, Theorem 4.4.17] and the lateral class $\mathcal{C}(\mathbf{Y})\mathbf{S}$ has dimension at least k . Therefore, it follows that the rank of the linear system $[\mathbf{I}_k \otimes \mathbf{X}^\top - \mathbf{Y} \otimes \mathbf{I}_k] \mathbf{x} = \mathbf{0}$ is smaller than or equal to $k^2 - k$. $\square$

Proposition 2. Consider an instance $\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3 \in \mathbb{F}_q^{k \times mn}$ of the 2-sMCE problem such that $\mathbf{G}_2 \mathbf{G}_3^\top, \mathbf{G}_0 \mathbf{G}_3^\top, \mathbf{G}_1 \mathbf{G}_2^\top \in \mathbb{F}_q^{k \times k}$ are full rank matrices. Write $\mathbf{X} = (\mathbf{G}_2 \mathbf{G}_3^\top)(\mathbf{G}_2 \mathbf{G}_1^\top)^{-1}$ and $\mathbf{Y} = (\mathbf{G}_3 \mathbf{G}_2^\top)(\mathbf{G}_3 \mathbf{G}_0^\top)^{-1}$. Then

$$\mathbf{S}\mathbf{X} = \mathbf{Y}\mathbf{S} \quad (5)$$

holds. Moreover, $\mathbf{x} = \text{vec}(\mathbf{S})$ is a nonzero solution of the linear system $\mathbf{M}\mathbf{x} = \mathbf{0}$ where

$$\mathbf{M} = [\mathbf{I}_k \otimes \mathbf{X}^\top - \mathbf{Y} \otimes \mathbf{I}_k]$$

has a rank at most equal to $k^2 - k$.

Proof. The proof is essentially the same as in Proposition 1 with the difference we derive Equation (5) from Equations (2) and (3). $\square$

Remark 3. For random instances $\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3 \in \mathbb{F}_q^{k \times mn}$ of the 2-sMCE problem, the rank-maximality condition on $\mathbf{G}_0 \mathbf{G}_1^\top, \mathbf{G}_2 \mathbf{G}_3^\top, \mathbf{G}_0 \mathbf{G}_3^\top$ and $\mathbf{G}_1 \mathbf{G}_2^\top$ is expected to hold. In addition, the linear system given by

$$\begin{bmatrix} \mathbf{I}_k \otimes \mathbf{X}_1^\top - \mathbf{Y}_1 \otimes \mathbf{I}_k \\ \mathbf{I}_k \otimes \mathbf{X}_2^\top - \mathbf{Y}_2 \otimes \mathbf{I}_k \end{bmatrix} \mathbf{x} = \mathbf{0}, \quad (6)$$

where $(\mathbf{X}_1, \mathbf{Y}_1)$ and $(\mathbf{X}_2, \mathbf{Y}_2)$ are the matrices given in Propositions 1 and 2, is expected to have a rank equal to $k^2 - 1$. This is because the centralizers in Propositions 1 and 2 are expected to trivially intersect for random matrices, and hence we expect in practice that the space of solutions for the linear system to collapse to a one-dimensional kernel space.

3.2 On the Details of Step 2 in Figure 1

Once we recover matrix $\mathbf{S} \in \text{GL}_k(\mathbb{F}_q)$ by solving the linear system in Equation (6), we proceed by determining the (skew) symmetric matrices $\mathbf{A} \in \mathbb{S}_m(\mathbb{F}_q)$ and $\mathbf{B} \in \mathbb{S}_n(\mathbb{F}_q)$ as follows. Notice that

$$(\mathbf{S}^{-1}\mathbf{G}_1) = \mathbf{G}_0(\mathbf{A}^\top \otimes \mathbf{B}) \Leftrightarrow (\mathbf{S}^{-1}\mathbf{G}_1)(\mathbf{A}^{-\top} \otimes \mathbf{I}_n) = \mathbf{G}_0(\mathbf{I}_m \otimes \mathbf{B}),$$

which determines a linear system of kmn equations in the $(m^2 + n^2)$ variables $\mathbf{X} = \mathbf{A}^{-\top}$ and $\mathbf{Y} = \mathbf{B}$ and is expected to have a rank equal to $\min\{kmn, m^2 + n^2 - 1\}$. Moreover, if $\delta = \frac{k}{mn}$ denotes the code rate, we find that if $kmn = \delta m^2 n^2 \geq n^2 + m^2$, then the rank of the above system becomes $m^2 + n^2 - 1$ for values of $\delta \geq \frac{1}{n^2} + \frac{1}{m^2}$.

Remark 4. Notice that we can consider double of equations by including the linear relations determined by $\mathbf{G}_2$ and $\mathbf{G}_3$; that is,

$$(\mathbf{S}^{-1}\mathbf{G}_3) = \mathbf{G}_2(\mathbf{A}^\top \otimes \mathbf{B}) \Leftrightarrow (\mathbf{S}^{-1}\mathbf{G}_3)(\mathbf{A}^{-\top} \otimes \mathbf{I}_n) = \mathbf{G}_2(\mathbf{I}_m \otimes \mathbf{B}).$$

Furthermore, exploiting the (skew) symmetric structure in $\mathbf{A}$ and $\mathbf{B}$ allows us to reduce the number of variables from $m^2 + n^2$ to $\frac{1}{2}m(m - \epsilon_A) + \frac{1}{2}n(n - \epsilon_B)$, where $\mathbf{A}^\top = \epsilon_A \mathbf{A}$ and $\mathbf{B}^\top = \epsilon_B \mathbf{B}$ for some $\epsilon_A, \epsilon_B \in \{1, -1\}$. These observations suggest a double of equations and around half of variables, implying a wider interval of values for δ where the linear system has a one-dimensional right kernel, allowing a successful recovery of matrices $\mathbf{A}$ and $\mathbf{B}$.

4 Cryptographic Implications

To have a better picture of the practical implications of our results in Section 3, we focus our experiments on the standard setting $k = m = n$ as in [CNP⁺23, BBMP24]. We note that the recent (and concurrent) work [GMPT26] also analyzes the security of the blind signature scheme in [KLP26]. However, their conclusions still illustrate an exponential time complexity while our results and experiments illustrate a polynomial time complexity for the same parameter sets.

Notice that for the scenario $k = m = n$, we have $\frac{1}{n} = \frac{k}{mn} \geq \frac{1}{n^2} + \frac{1}{m^2} = \frac{2}{n^2}$ for any integer $n \geq 2$; therefore, our algorithm in Figure 1 is expected to solve random instances of 2-sMCE problem. To illustrate the practicability of algorithm in Figure 1, we land our experiments for an increased variant of the MEDS parameter sets as analyzed in [GMPT26]: they scale the matrix dimensions by a factor of $\sqrt{2}$. In particular, the authors in [GMPT26] suggest a complexity of 72 bits for the smallest parameter set. We also provide experiments for the parameter set proposed in [KLP26], which were estimated to provide 128-bits of security. We show in Table 1 that we can solve the 2-sMCE instances for these parameter sets in a matter of minutes.¹ Therefore, our results implies that the blind signature construction in [KLP26] can be broken in polynomial time. We provide a SageMath implementation of our algorithm in [Com26].

Table 1: Experiments for solving the 2-sMCE for $k = m = n$ over $\mathbb{F}_q$.

	Dimension (n)	Field size (q)	Time
[GMPT26]	20	4093	7.4 sec
	31	4093	2.2 min
	42	2039	17.6 min
[KLP26]	46	4093	33.7 min

¹Our experiments also illustrate that sMCE is solvable in around the same amount of time as 2-sMCE.

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